

CRADLE: Transient Security–Dependability Analysis of Crypto-Ransomware Kill-Chains in MEC Micro-Datacenters under Footprint-Aware Detection and Backup–Restore Defense

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Abstract—Multi-access edge computing (MEC) micro-datacenters place a handful of virtualized servers at the network edge, where they are exposed to crypto-ransomware campaigns that move laterally, steal data, and then encrypt it to deny service. Unlike a classical breach, a ransomware incident is survivable: locked servers can be restored from immutable backups, so its true cost is the transient loss of confidentiality, service, and data recency accrued during the race between the attacker’s kill-chain and the defender’s detect-and-restore response. We develop CRADLE, a continuous-time Markov reward model, drawn as a stochastic reward net, of an ordered compromise–exfiltrate–encrypt–lock kill-chain on a MEC micro-datacenter, defended by an endpoint monitor whose detection hazard grows with the encryption and exfiltration footprint and by an isolate-and-restore response with explicit recovery-point and recovery-time semantics. We derive transient state probabilities, accumulated-reward security losses, first-passage detection and resolution distributions, and closed per-incident expectations via the fundamental matrix, and we prove the footprint-aware detection breaks the memoryless recovery law of prior uniform-recovery models. A defender co-design procedure jointly tunes detection sensitivity and backup cadence to minimize expected total cost, tracing a security-versus-spend Pareto frontier. An independent Gillespie simulation validates every analytic result. The study yields actionable, and at times counterintuitive, guidance for edge ransomware resilience.

Index Terms—Multi-access edge computing, ransomware, kill-chain, stochastic reward nets, continuous-time Markov chains, transient analysis, dependability, security, backup and restore, moving-target defense, performability.

I. INTRODUCTION

MULTI-ACCESS edge computing (MEC) pushes computation to micro-datacenters (MEDCs) of a few virtualized servers placed beside base stations, delivering the low latency that immersive, industrial and vehicular applications demand [1], [2]. That proximity is also an attack surface: an MEDC sits in a physically and administratively exposed location, hosts multi-tenant virtual machines and containers, and is a natural staging point for adversaries seeking the data and control flows that traverse the edge [3]. The dom-

inant monetized threat against such fleets today is *crypto-ransomware*: an adversary gains an initial foothold, moves laterally between servers, exfiltrates data for double-extortion leverage, and then encrypts data at rest to deny service until a ransom is paid [4], [5].

Quantitative stochastic models have proven invaluable for reasoning about the security and dependability of such systems before deployment. The MEDC-under-attack model of Liu *et al.* [7] is the closest antecedent: it analyses a lateral-movement campaign on an MEDC as an absorbing continuous-time Markov chain (CTMC) and computes transient infection, exfiltration and loss metrics. That model, however, makes two simplifications that matter for ransomware. First, it treats *recovery* as a single uniform rate to an absorbing “defense-deployed” state, independent of how far the attack has progressed—so the recovery probability is exactly memoryless. In reality, detection is driven by the attacker’s own noise: encryption produces a high-entropy I/O storm and mass data egress that an endpoint monitor sees precisely *because* the campaign is advancing. Second, it has no notion of *data-at-rest denial and its reversal*: encryption is not modelled as a recoverable service-denial state, and there is no backup-and-restore layer with recovery-point (RPO, how much data is lost) or recovery-time (RTO, how long restoration takes) semantics—exactly the levers a defender actually controls.

This paper closes that gap. We propose **CRADLE** (*Crypto-Ransomware Attack: Detection-race, Loss and restore-Economics*), a transient Markov reward model, expressed as a stochastic reward net (SRN), of a crypto-ransomware double-extortion kill-chain on an MEDC, defended by a footprint-aware endpoint detection-and-response (EDR) monitor and an immutable-backup isolate-and-restore response. A ransomware incident in CRADLE is *survivable*: every trajectory is eventually restored, and the object of study is the transient security and dependability loss—confidentiality, service availability, data recency, and downtime—accumulated while the kill-chain races the response.

Our contributions are:

- An **ordered double-extortion ransomware SRN** (Section IV) whose per-server kill-chain is *Vulnerable*→*Compromised*→*Exfiltrating*→*Encrypting*→*Locked*, with lateral movement scaling in the active footprint, and in which encryption is a

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Reproducibility: all models, solvers, the simulator, and the scripts that regenerate every figure, table and in-text number are openly available at <https://github.com/anhnt2407/504-CRADLE-Transient-Ransomware-Kill-Chain-Detection-Backup-Restore-MEC-Micro-Datacenter-SRN>.

recoverable service-denial state rather than an absorbing breach.

- **A footprint-aware detection hazard** keyed on the encryption/exfiltration I/O footprint, coupled to an **isolate-and-restore response with explicit RPO/RTO** and imperfect restore coverage (Section III–IV). We prove that this coupling breaks the memoryless recovery law of uniform-recovery models: the resolution curve is S-shaped, not exponential.
- **Exact transient and per-incident measures** (Section V): transient probabilities and accumulated-reward losses via matrix exponentials, first-passage detection and resolution distributions, and closed per-incident expectations (mean time-to-detection, mean time-to-resolution, expected datasets exfiltrated, denial server-days, and data loss at restore) via the fundamental matrix.
- **A defender co-design algorithm** (Section VI) that jointly optimizes detection sensitivity and backup cadence to minimize expected total cost, tracing the security-versus-operational-spend Pareto frontier and quantifying the penalty of a detection-blind backup schedule.
- **A dual analytic/simulation validation** (Section VII) and a numerical study (Section VIII) that surface actionable design guidance, including the counterintuitive result that sharper detection lets a defender back up *less* often.

II. RELATED WORK

Stochastic dependability and security modelling. Markov reward models, stochastic Petri/reward nets and semi-Markov processes are the standard vehicles for quantifying availability, reliability and security of computing systems [8], [9], [10], [11]. Trivedi and co-workers pioneered the unified treatment of performance, dependability and security (“performability” and quantitative security) that this paper follows. The dependability/security taxonomy of Avižienis *et al.* [25] frames encryption-based denial as a combined integrity-and-availability fault whose handling (detection, isolation, recovery) is precisely what CRADLE quantifies. Intrusion-tolerance models in this lineage—for instance the state-transition security model of Madan *et al.* [24], which derives mean time to security failure from an absorbing CTMC—established the reward- and first-passage-based measures that we lift here from a single host to a footprint-coupled MEDC fleet with data-centric restore.

MEDC and edge under attack. Liu *et al.* [7] model a lateral-movement attack on an MEDC as an absorbing CTMC with a uniform recovery to a “good” state; it is our direct antecedent and baseline. A subsequent line models *self-propagating* attacks on micro-segmented edge fabrics with two competing absorbing outcomes (contained vs. irreversibly breached) and detection hazards that grow with the compromised-node count [3]. CRADLE differs in threat model and objective: our detection hazard is keyed on the *encryption/exfiltration* footprint rather than compromised-node count; encryption is a *recoverable* denial state, not a terminal breach; and our response is a backup *restore* with RPO/RTO data-loss economics rather than a re-image.

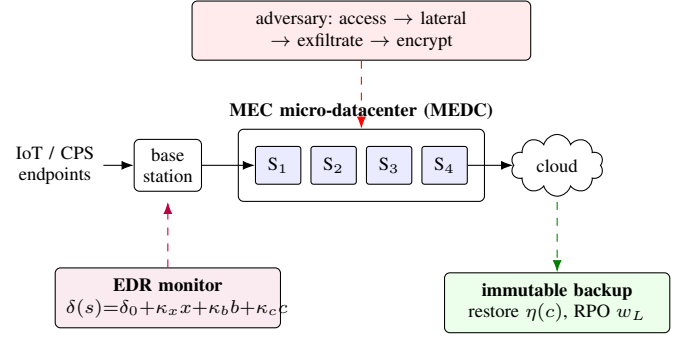


Fig. 1. CRADLE system architecture: an MEDC of m MEC servers, a footprint-aware EDR monitor, and an immutable-backup isolate-and-restore plane, under a crypto-ransomware kill-chain.

Ransomware analysis. Empirical and systems work dissects ransomware behaviour and defenses—file-system self-healing, decoy monitoring and behavioural detection [4], [5], [6]. These are measurement/mechanism studies; to our knowledge no prior transient Markov reward model captures the exfiltrate-then-encrypt kill-chain together with detection and backup-restore economics.

Moving-target defense and proactive recovery. MTD (IP shuffling, VM migration, re-imaging) and proactive/rejuvenation-based recovery reset attacker progress on a timer or a trigger [12], [13]. Our restore is conceptually a triggered reset, but it is *data-centric*: what is restored is encrypted data from an immutable backup, and the residual cost is the recovery point, an aspect these works do not model.

Numerical methods. We use the matrix exponential for transient solutions, Jensen’s uniformization as an independent cross-check [14], a Van Loan-style augmented exponential for accumulated rewards [15], [16], the fundamental matrix for per-incident expectations, and Gillespie’s exact stochastic simulation for validation [17].

III. SYSTEM ARCHITECTURE AND THREAT MODEL

A. Architecture

Fig. 1 depicts the system. An MEDC hosts m MEC servers behind one or more base stations, each running a virtualization stack (hypervisor or container engine) that instantiates tenant services close to endpoint and IoT/CPS devices. Two defensive planes overlay the compute fabric. An *EDR monitor* continuously observes host and network telemetry; its alert hazard rises with the volume of ransomware-indicative activity—high-entropy write bursts from encryption and anomalous data egress from exfiltration. An *immutable-backup* subsystem periodically snapshots server data to write-once storage; upon an incident it drives an *isolate-and-restore* response that network-isolates the MEDC (halting lateral movement and exfiltration) and restores servers from the last good snapshot.

B. Threat model

The adversary follows a ransomware double-extortion kill-chain. It obtains initial access to one server, then propagates

TABLE I
STATE ENCODING OF THE CRADLE SRN.

symbol	meaning	service
$V = m - p$	Vulnerable (clean, serving)	up
a	Compromised (foothold, staging)	up
x	Exfiltrating (data egress in progress)	up (leaking)
b	Encrypting (crypto in progress)	degraded
c	Locked (data denied)	down
d	phase: 0 stealth / 1 response	—
GOOD	restored (absorbing)	up

laterally (the more servers it controls, the faster it spreads). On a controlled server it *exfiltrates* data (confidentiality loss and extortion leverage) and only *then encrypts* it, completing to a *locked* state that denies service. The defender does not know it is under attack until the EDR fires; detection is imperfect and its timeliness depends on the attacker's footprint. We assume network isolation is effective once triggered (halting further spread and egress), that in-flight host-local encryption may still complete before restore, and that immutable backups are not themselves corruptible (a design goal of modern write-once backup). We do not model the ransom-payment decision: CRADLE is a defender-side resilience model whose premise is restore-not-pay.

IV. THE CRADLE STOCHASTIC REWARD NET

A. State space

The MEDC has m servers, each in exactly one kill-chain stage. A transient (operational) state is the tuple

$$s = (a, x, b, c, d), \quad (1)$$

where a, x, b, c count servers that are *Compromised*, *Exfiltrating*, *Encrypting* and *Locked*, and $d \in \{0, 1\}$ is the campaign phase: $d=0$ (stealth/undetected) or $d=1$ (response/contained). The number of clean *Vulnerable* servers is $V = m - p$ with footprint $p = a + x + b + c$, and the *active* (non-locked) foothold is $h = a + x + b$. Validity requires $a, x, b, c \geq 0$ and $p \leq m$; the stealth phase exists for every lattice point, the response phase only for $p \geq 1$ (a campaign must exist to be responded to). A single absorbing state GOOD represents the fully restored, incident-resolved MEDC. Table I summarizes the encoding. The lattice has $\binom{m+4}{4}$ points per phase, giving 139 transient states +1 for the base case $m = 4$.

B. Transition rules

Fig. 2 shows the kill-chain and phase automaton; the aggregate rate laws (verified arc-by-arc against the reachability graph) are in Table II. In the stealth phase, a clean server is first compromised by lateral movement at a rate proportional to the active foothold h (epidemic spread), inheriting the footprint-scaling of [7]; a compromised server begins exfiltration at rate ϵ each; an exfiltrating server completes (one dataset stolen) and *then* begins encryption at rate ϕ ; an encrypting server completes to Locked at rate ρ . The campaign is detected—moving to the response phase—at the footprint-aware hazard

$$\delta(s) = \delta_0 + \kappa_a a + \kappa_x x + \kappa_b b + \kappa_c c, \quad \delta(s) = 0 \text{ if } p=0, \quad (2)$$

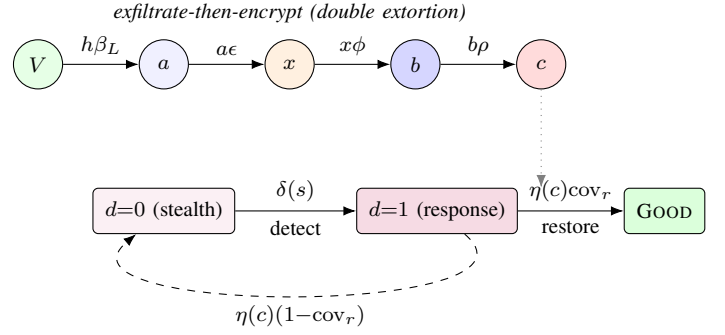


Fig. 2. CRADLE kill-chain (top: per-server stages with aggregate rates) and campaign-phase automaton (bottom: stealth $\rightarrow$ response $\rightarrow$ restored). The detection hazard $\delta(s)$ grows with the encryption/exfiltration footprint; the restore rate $\eta(c)$ falls with the number of locked servers.

TABLE II
TRANSITION (RATE-LAW) TABLE OF THE CRADLE SRN.

activity	effect on (a, x, b, c, d)	rate	guard
initial access	$(0, 0, 0, 0, 0) \rightarrow (1, 0, 0, 0, 0)$	β	$p=0, V \geq 1$
lateral move	$a \rightarrow a+1$	$h\beta_L$	$V \geq 1, h \geq 1$
exfil. onset	$a \rightarrow a-1, x \rightarrow x+1$	$a\epsilon$	$a \geq 1, d=0$
exfil. done	$x \rightarrow x-1, b \rightarrow b+1$	$x\phi$	$x \geq 1, d=0$
encrypt done	$b \rightarrow b-1, c \rightarrow c+1$	$b\rho$	$b \geq 1$
detection	$d: 0 \rightarrow 1$	$\delta(s)$	$p \geq 1, d=0$
restore	$\rightarrow \text{GOOD}$	$\eta(c)\text{cov}_r$	$d=1$
restore (residual)	$\rightarrow (1, 0, 0, 0, 0)$	$\eta(c)(1-\text{cov}_r)$	$d=1$

with $\kappa_c > \kappa_b > \kappa_x \geq \kappa_a$: locked servers raise user-visible alarms, encryption is a noisy I/O storm, exfiltration leaves an egress signature, and a bare foothold is nearly silent. In the response phase, network isolation halts lateral movement, exfiltration and new encryption onsets; host-local encryption already running still completes ($b \rightarrow c$ at ρ), and the restore fires to GOOD at the state-dependent rate

$$\eta(c) = \frac{1}{\text{RTO}_0 + \text{RTO}_1 c}, \quad (3)$$

i.e. the recovery time grows with the number of locked servers to be restored. With restore coverage $\text{cov}_r \in (0, 1]$, a fraction $1 - \text{cov}_r$ of restores are incomplete and leave a residual foothold, returning the campaign to $(1, 0, 0, 0, 0)$.

C. Generator

Let $\mathcal{S}$ be the transient state set and GOOD the absorbing state. The infinitesimal generator Q has off-diagonal entries $q_{s,s'}$ equal to the sum of rates of arcs $s \rightarrow s'$ and diagonal $q_{s,s} = -\sum_{s' \neq s} q_{s,s'}$; the GOOD row is zero. Algorithm 1 synthesizes Q from the rate laws; its rows sum to zero by construction, which we verify numerically. Default parameters (per day) are listed in Table III; they place the incident on a multi-day timescale in which detection, encryption and restore genuinely compete.

V. SECURITY-DEPENDABILITY MEASURES

Let π_0 be the initial distribution (all mass on the clean state $(0, 0, 0, 0, 0)$) and $\pi(t) = \pi_0 e^{Qt}$ the transient distribution.

TABLE III
DEFAULT PARAMETERS (TIME UNIT: DAYS).

symbol	meaning	default	unit
m	MEC servers in the MEDC	4	servers
β	initial-access rate	0.50	d^{-1}
β_L	lateral-movement rate / active foothold	0.40	d^{-1}
ϵ	exfiltration-onset rate / compromised	0.80	d^{-1}
ϕ	exfiltration-completion rate / exfiltrating	1.00	d^{-1}
ρ	encryption-completion rate / encrypting	2.00	d^{-1}
δ_0	baseline detection hazard	0.20	d^{-1}
κ_x	detection gain / exfiltrating server	0.50	d^{-1}
κ_b	detection gain / encrypting server	1.50	d^{-1}
κ_c	detection gain / locked server	0.80	d^{-1}
RTO_0	base restore time	0.25	d
RTO_1	restore-time growth / locked server	0.15	d
w_L	data loss / locked server (RPO)	1.00	unit
cov_r	restore coverage	1.00	–

Transient marginals. The expected number of servers in a stage is a reward-weighted marginal, e.g. $\mathbb{E}[c(t)] = \sum_s c_s \pi_s(t)$; the expected available service fraction is $A(t) = 1 - \mathbb{E}[c(t)]/m$, and the probability of service denial is $\mathbb{P}(c(t) \geq 1)$.

Accumulated-reward losses. For a reward-rate vector r , the expected accumulated reward to time t is $L_r(t) = \int_0^t r^\top \pi(u) du$, obtained without quadrature from the augmented exponential

$$\begin{bmatrix} \pi(t) \\ \mathbf{A}(t) \end{bmatrix} = \exp \left(\begin{bmatrix} Q^\top & 0 \\ I & 0 \end{bmatrix} t \right) \begin{bmatrix} \pi_0 \\ 0 \end{bmatrix}, \quad L_r(t) = r^\top \mathbf{A}(t). \quad (4)$$

We instantiate r as the locked count c (service-denial server-days), the exfiltration-completion rate ϕx restricted to stealth (datasets stolen), and the restore-flux data loss $\eta(c)w_L c$ restricted to response (data lost at restore, an RPO quantity).

First-passage distributions. Because a trajectory is stealthy exactly until detection, the detection CDF is $\mathbb{P}(\text{detected} \leq t) = 1 - \sum_{s:d_s=0} \pi_s(t)$, and the resolution (restore) CDF is $\mathbb{P}(\text{resolved} \leq t) = \pi_{\text{GOOD}}(t)$.

Per-incident expectations (fundamental matrix). Let U be Q restricted to $\mathcal{S}$. The expected sojourn vector before absorption is

$$\tau^\top = \pi_{0,\mathcal{S}}^\top (-U)^{-1}, \quad (5)$$

from which every per-incident total is the linear functional $\tau^\top r$: mean time-to-detection $\text{MTTD} = \sum_{d_s=0} \tau_s$, mean response duration $\sum_{d_s=1} \tau_s$, mean time-to-resolution $\text{MTTR} = \sum_s \tau_s$, expected datasets exfiltrated $\phi \sum_{d_s=0} x_s \tau_s$, expected denial server-days $\sum_s c_s \tau_s$, and expected data loss $w_L \sum_{d_s=1} \eta(c_s) c_s \tau_s$. The identity $\text{MTTR} = \text{MTTD} + \mathbb{E}[\text{response}]$ holds exactly.

Structural result (non-memoryless recovery). In a uniform-recovery model where every transient state exits to the absorbing state at a single rate γ , the resolution CDF is exactly $1 - e^{-\gamma t}$. CRADLE is qualitatively different, as the following makes precise.

Lemma 1 (Non-memoryless resolution). *For any parameters with $\delta_0 \geq 0$ and $\eta(\cdot) < \infty$, the time-to-resolution T_R (first passage from the clean state to GOOD) is a phase-type random variable with density $f_{T_R}(0) = 0$. In particular T_R is not*

Algorithm 1 CRADLE generator synthesis

Require: parameters θ , size m

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1: enumerate  $\mathcal{S} = \{(a, x, b, c, d) : a+x+b+c \leq m, d \in \{0, 1\}, (d=1 \Rightarrow p \geq 1)\}$ 
2:  $Q \leftarrow 0$ 
3: for all  $s = (a, x, b, c, d) \in \mathcal{S}$  do
4:   for all  $(s', \lambda)$  in  $\text{TRANSITIONS}(s; \theta)$  (Table II) do
5:      $Q_{s,s'} \leftarrow Q_{s,s'} + \lambda$ ;  $Q_{s,s} \leftarrow Q_{s,s} - \lambda$ 
6:   end for
7: end for
8: assert  $\max_s |\sum_{s'} Q_{s,s'}| < \varepsilon$ 
9: return  $Q$ 
```

exponentially distributed, and the resolution CDF is not of the form $1 - e^{-\gamma t}$ for any $\gamma > 0$.

Proof. Absorption into GOOD occurs only from response states ($d=1$), which are reachable only through the detection event ($d: 0 \rightarrow 1$); no arc leads from a stealth state directly to GOOD. Hence every path to absorption visits at least two distinct transient states, so $T_R \geq T_1 + T_2$ with T_1 the (positive) time to detection and T_2 the (positive) residual to restore. As a first passage in a finite CTMC starting from a single transient state, T_R is phase-type; because at least two phases are traversed with probability one, its density at the origin vanishes, $f_{T_R}(0^+) = 0$. The exponential density is $\gamma e^{-\gamma t}$ with $f(0^+) = \gamma > 0$, so T_R cannot be exponential. $\square$

The stealth window thus imprints an initial “dead time” on the resolution curve, making it S-shaped; the coupling of detection to attack progress is exactly what a uniform-recovery model [7] cannot represent. Section VIII quantifies the departure.

Security-dependability cost. We weight the incident harms into a single security-dependability loss and add the defender’s operational spend:

$$S = c_{br} \mathbb{E}[\text{exfil}] + c_{dn} \mathbb{E}[\text{denial}] + c_{dl} \mathbb{E}[\text{loss}] + c_{rt} \mathbb{E}[\text{response}], \quad (6)$$

$$J = S + \underbrace{C_{\text{mon}}(\sigma_D) + C_{\text{bkp}}(\tau_b)}_O, \quad (7)$$

where σ_D and τ_b are the defender levers introduced next.

VI. SOLUTION ALGORITHMS AND DEFENDER CO-DESIGN

A. Generator synthesis and exact solution

Algorithm 1 builds Q ; the transient distribution and accumulated rewards follow from (4) by forming a single base-step propagator $e^{Q\Delta t}$ and composing it along a uniform grid (exact, no step error). The per-incident totals follow from the one linear solve (5). We cross-check the matrix exponential against Jensen’s uniformization.

B. Defender co-design

The defender holds two orthogonal levers. *Detection sensitivity* $\sigma_D \geq 1$ scales the footprint coefficients $(\kappa_x, \kappa_b, \kappa_c) \mapsto$

Algorithm 2 Detection/backup co-design

Require: base parameters θ , cost config, lever boxes $[\sigma_{lo}, \sigma_{hi}], [\tau_{lo}, \tau_{hi}]$

- 1: evaluate J on a $\sigma_D \times \tau_b$ grid via Alg. 1+(5); set $(\sigma_D, \tau_b) \leftarrow$ grid minimizer
- 2: **repeat**
- 3: $\tau_b \leftarrow \text{GOLDENMIN}_{\tau} J(\sigma_D, \cdot)$
- 4: $\sigma_D \leftarrow \text{GOLDENMIN}_{\sigma} J(\cdot, \tau_b)$
- 5: **until** converged
- 6: **return** (σ_D^*, τ_b^*) , J^* , and the non-dominated frontier $\{(O, S)\}$

TABLE IV

ANALYTIC VS. GILLESPIE SIMULATION (200,000 TRAJECTORIES).

quantity	analytic	simulation (95% CI)	agree
E[datasets exfiltrated]	0.5657	0.5642 ± 0.0028	✓
E[denial] (srv-days)	0.3894	0.3854 ± 0.0037	✓
E[data loss] (RPO)	0.3823	0.3801 ± 0.0025	✓
MTTD (days)	3.6366	3.6308 ± 0.0101	✓
MTTResolve (days)	3.9440	3.9369 ± 0.0103	✓

$\sigma_D(\kappa_x, \kappa_b, \kappa_c)$ —a sharper EDR shortens the stealth window but costs more to run, $C_{\text{mon}}(\sigma_D) = c_{\text{mon}}\sigma_D$. *Backup cadence* τ_b (hours) sets the recovery point: the data loss per locked server scales with τ_b (i.e. $w_L \propto \tau_b$), while the backup/storage overhead scales with the frequency, $C_{\text{bkp}}(\tau_b) = c_{\text{bkp}}\tau_{b0}/\tau_b$. Algorithm 2 minimizes $J(\sigma_D, \tau_b)$ of (7): it maps the surface on a grid, then polishes with alternating one-dimensional golden-section searches (each objective evaluation is one solve (5)). Sweeping the levers and retaining non-dominated (O, S) pairs yields the Pareto frontier.

C. Independent simulation

Algorithm-agnostic validation is provided by an exact Gillespie simulation [17] that consumes only the per-state transition list, evolves each trajectory clean-to-restored, and rebuilds the transient probabilities, CDFs and per-incident totals with confidence intervals.

VII. SIMULATION VALIDATION

We validate the analytic engine three ways (a “verification triangle”): (i) the matrix exponential against Jensen uniformization (agreement to solver tolerance, $\sim 10^{-12}$); (ii) the fundamental-matrix per-incident totals against the time-integrated accumulated reward of (4) (agreement to $\sim 10^{-15}$); and (iii) an independent Gillespie simulation of 200,000 trajectories. Table IV reports the third leg: every per-incident expectation agrees within the 95% confidence interval, as do the detection and resolution CDFs (maximum discrepancy < 0.01). All seventeen automated checks pass.

VIII. NUMERICAL RESULTS AND ANALYSIS

Unless stated otherwise, results use the defaults of Table III ($m = 4$) over a 20-day horizon. Analytic curves are solid; independent simulation estimates are markers.

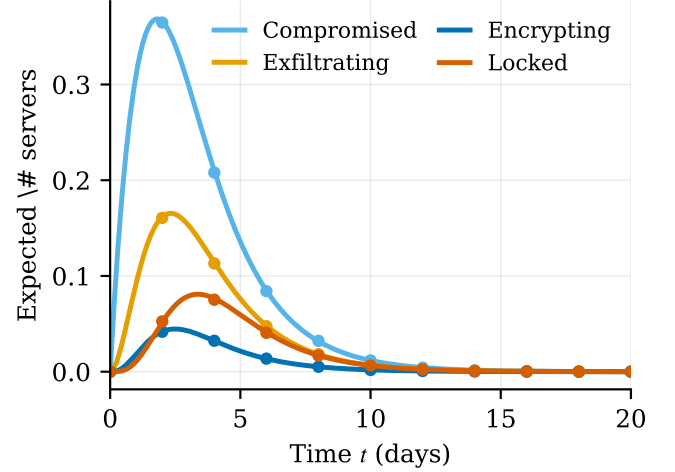


Fig. 3. Expected number of servers per kill-chain stage over time. Markers: Gillespie simulation.

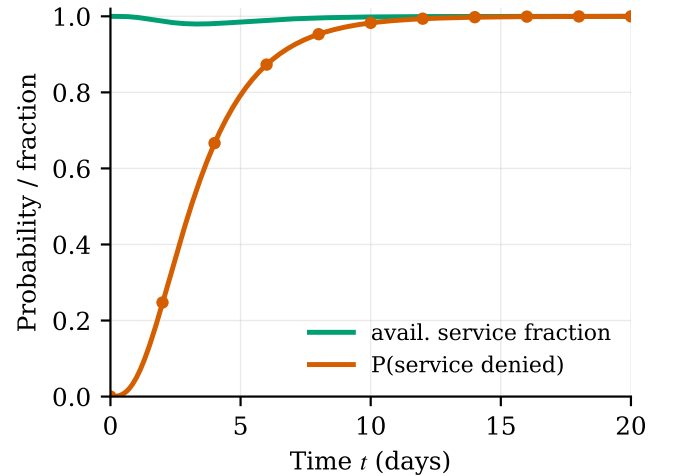


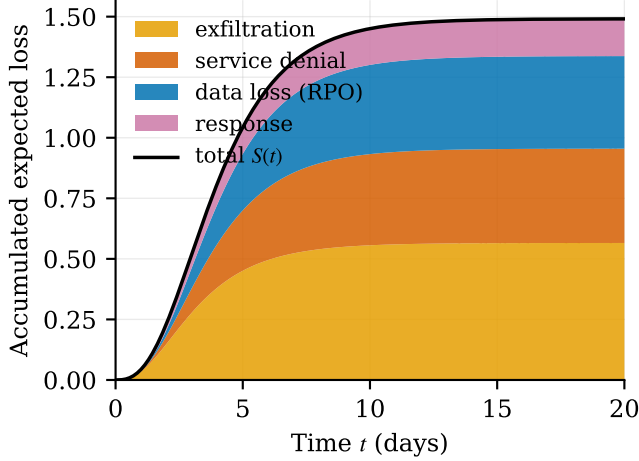
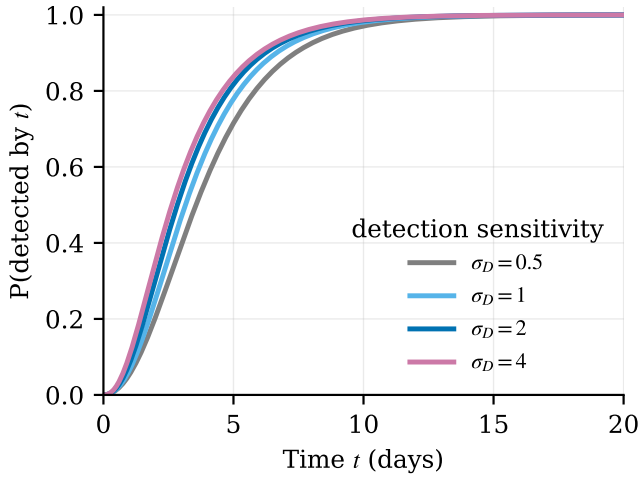
Fig. 4. Expected available service fraction and probability of service denial.

A. Kill-chain dynamics and service denial

Fig. 3 tracks the expected number of servers in each kill-chain stage. The compromise and exfiltration populations rise first, feeding a delayed wave of encryption and then locking; all populations then collapse as detection triggers isolation and restore. Fig. 4 translates this into service terms: the expected available fraction dips during the encryption wave and the probability of service denial peaks before restoration returns the MEDC to full service. The transient nature is essential—a steady-state model would miss the peak denial entirely.

B. Accumulated security-dependability loss

Fig. 5 decomposes the accumulated expected loss $S(t)$ into exfiltration, service denial, RPO data loss and response components. The loss accrues steeply during the active phase and saturates once incidents resolve; its composition—confidentiality vs. availability vs. recency—tells a defender which harm dominates and thus which lever to pull.

Fig. 5. Accumulated expected security-dependability loss $S(t)$, decomposed.Fig. 6. Detection CDF vs. detection sensitivity σ_D : sharper detection compresses the stealth window.

C. The detection race and non-memoryless recovery

Fig. 6 shows the detection CDF for a range of detection sensitivities: sharper detection compresses the stealth window, and—because the hazard grows with the attacker’s own encryption footprint—even a modest sensitivity eventually detects every campaign. Fig. 7 is the headline structural result: the resolution curve is distinctly S-shaped, with an initial dead-time while the campaign is still stealthy, and departs from the best-fit memoryless exponential $1 - e^{-t/\text{MTTR}}$ by a wide margin (0.21 at its maximum). A uniform-recovery model of the form [7] cannot represent this behaviour; the coupling of detection to attack progress is precisely what makes edge ransomware resolution non-memoryless.

D. Defender co-design

Fig. 8 maps the expected total cost J over the detection-sensitivity/backup-cadence plane; it is convex-like with a clear interior basin. Cost rises steeply at short backup intervals

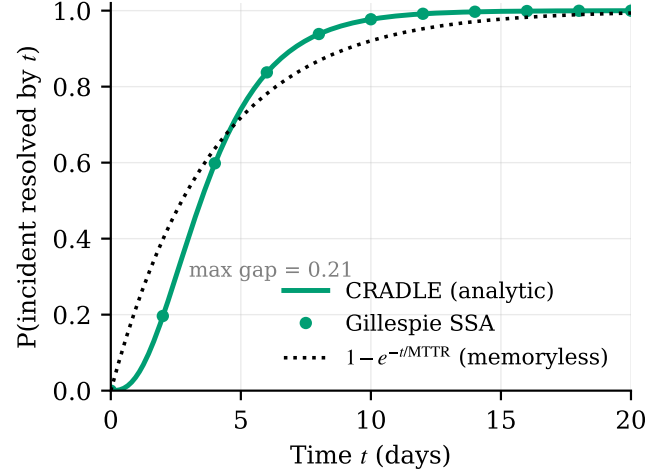
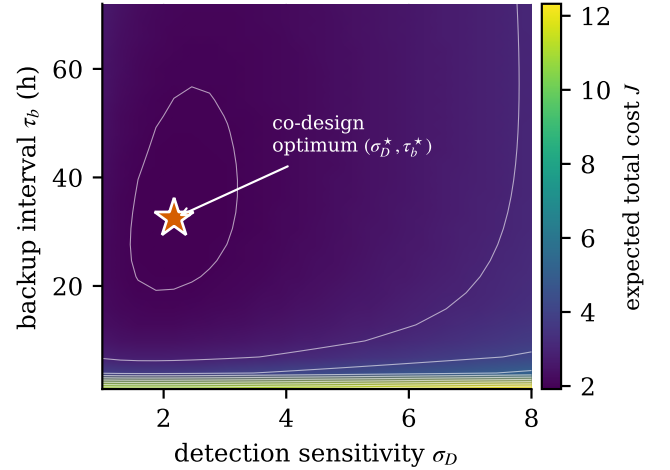


Fig. 7. Resolution (restore) probability: CRADLE is S-shaped and non-memoryless, unlike the exponential recovery of uniform-recovery models. Markers: simulation.

Fig. 8. Expected total cost $J(\sigma_D, \tau_b)$ with the co-design optimum.

(storage overhead) and at high sensitivity (monitoring overhead), while very loose backups or dull detection let the security loss blow up. Algorithm 2 locates the optimum $(\sigma_D^*, \tau_b^*) = (2.17, 32\text{h})$, where the security loss and operational spend are balanced ($S^* \approx O^*$). Fig. 9 traces the corresponding Pareto frontier; the min-total-cost design sits at its knee, where marginal security gain equals marginal spend. Table V contrasts the baseline, a detection-blind backup-only optimizer, and the co-design.

The counterintuitive finding is in the levers themselves (Figs. 10–11). A detection-blind operator, unable to reduce the number of servers that get locked, over-invests in *frequent* backups to fight data loss and still pays a penalty of 14% relative to the co-design. The co-designer instead sharpens detection so that far fewer servers ever reach the locked state—cutting expected denial by 62%—and can therefore *relax* the backup interval, saving storage overhead. *Better detection buys looser backups*: the two levers are economic substitutes, and

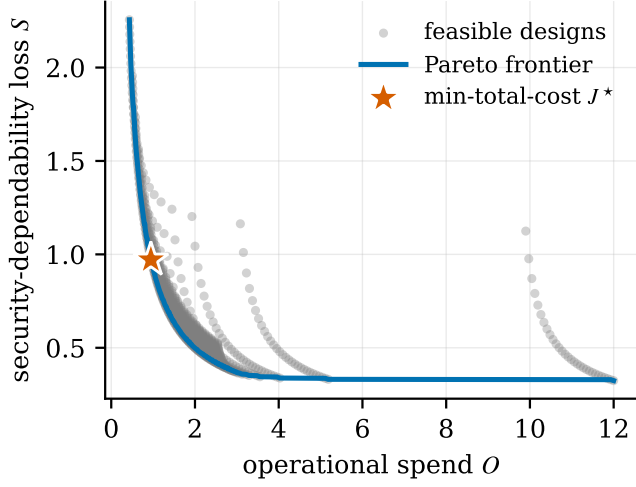


Fig. 9. Security-loss (S) vs. operational-spend (O) Pareto frontier; the star is the min-total-cost design.

TABLE V
BASELINE VS. DETECTION-BLIND VS. CO-DESIGN.

design	σ_D	τ_b (h)	E[exfil]	E[denial]	E[loss]	S	O
baseline	1.0	24	0.566	0.389	0.382	1.491	0.700
detection-blind	1.0	25	0.566	0.389	0.382	1.491	0.691
co-design *	2.17	32	0.385	0.149	0.296	0.971	0.947

treating them independently is strictly sub-optimal.

E. MEDC size and parameter sensitivity

Fig. 12 scales the MEDC from $m = 2$ to 6 servers (state space 29 to 419). The per-incident harms *saturate*: because detection resolves the incident before the campaign traverses the whole fleet, the expected datasets exfiltrated and denial server-days barely grow with size. The practical message is that, for edge micro-datacenters, detection latency and backup recency—not fleet size—govern ransomware risk. Fig. 13 ranks the drivers of expected data loss by scaled sensitivity: the backup recovery-point w_L dominates, followed by the encryption and exfiltration completion rates, while sharper detection coefficients reduce it—confirming the co-design intuition.

F. Robustness to imperfect restore

Real restores sometimes fail to fully evict the adversary—a re-imaged or restored server can be re-infected from a missed persistence foothold, and double-extortion crews often re-encrypt precisely to punish incomplete recovery. Fig. 14 sweeps the restore coverage cov_r from perfect down to 0.6. As coverage degrades, a fraction of incidents relapse into a fresh (smaller) campaign before finally resolving, and both the mean time-to-resolution and the expected data loss inflate super-linearly—by 21% and 43% respectively at $\text{cov}_r = 0.7$. The practical corollary is that restore *quality* (verified, immutable,

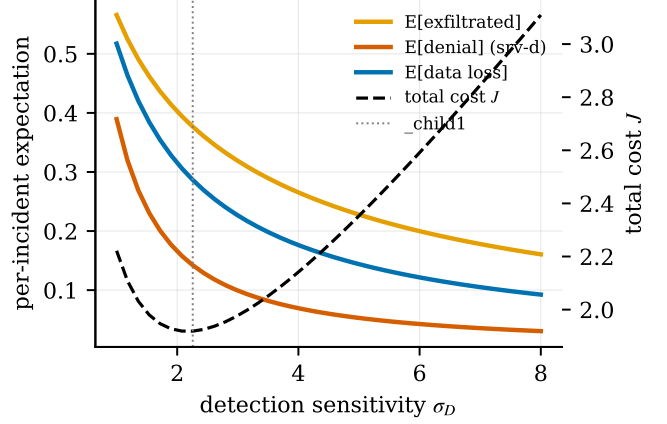


Fig. 10. Effect of detection sensitivity σ_D on the incident harms and on total cost J (dashed).

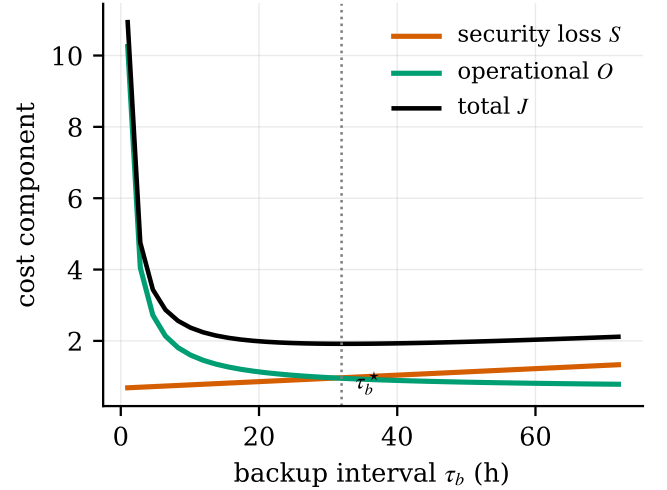


Fig. 11. Effect of backup interval τ_b : data-loss falls but operational cost rises as backups get more frequent; J has an interior optimum.

fully evicting) is not a second-order concern: a cheap-but-leaky restore can erase the benefit of an otherwise well-tuned detection/backup co-design.

G. Discussion and design guidance

Three guidelines follow. (1) *Instrument for the crypto footprint*. Because detection hazard rises with encryption I/O, tuning the EDR toward high-entropy write bursts yields the steepest security return. (2) *Co-design detection and backups*. They are substitutes; sharper detection prevents locking and lets backups be relaxed, so budgeting them separately overspends. (3) *Size is not the risk*. Per-incident harm saturates with fleet size, so consolidation for efficiency does not materially raise ransomware exposure once detection and restore are in place.

H. Threats to validity

CRADLE assumes exponential stage sojourns; heavy-tailed encryption or restore times would be better captured by a semi-

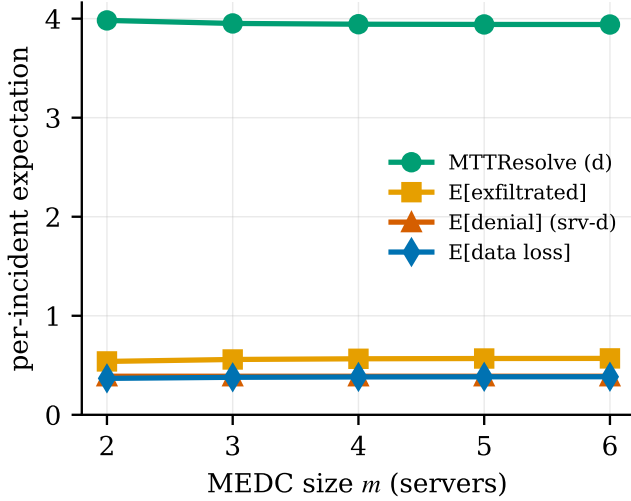


Fig. 12. Per-incident measures vs. MEDC size m : harms saturate with fleet size.

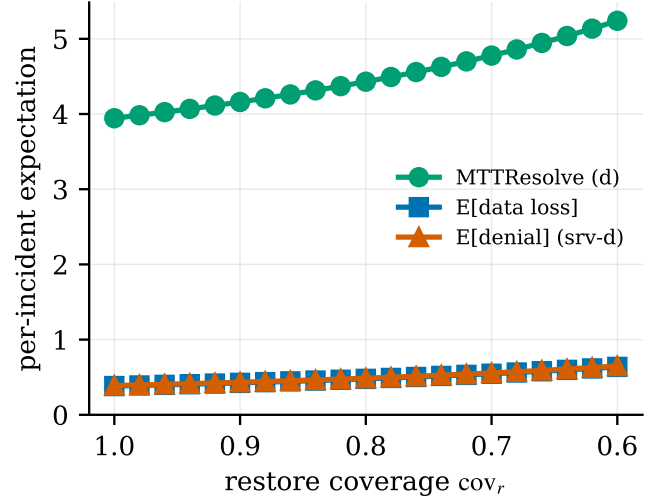


Fig. 14. Robustness to imperfect restore: as restore coverage cov_r degrades (right), relapse inflates resolution time and data loss super-linearly.

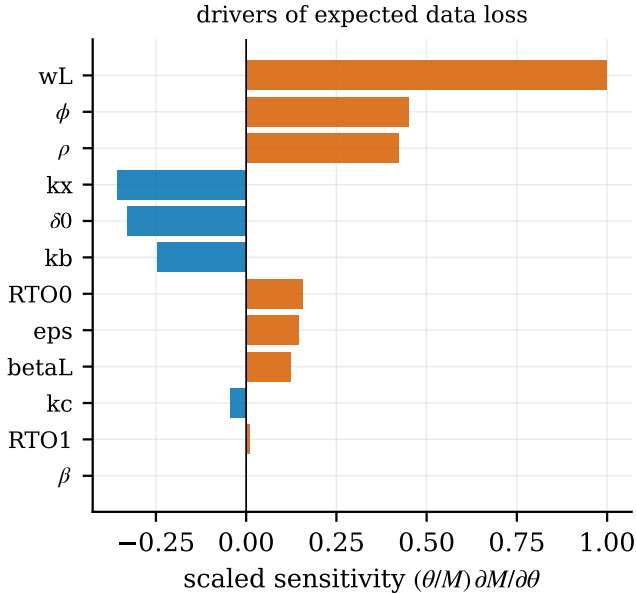


Fig. 13. Scaled sensitivity of expected data loss to model parameters.

Markov or phase-type refinement, which the accumulated-reward and fundamental-matrix machinery accommodates directly. We assume immutable backups are incorruptible and isolation is effective; relaxing these adds arcs but not new formalism. The cost weights are illustrative—the framework consumes any weights, and the qualitative guidance (substitutability of detection and backups, saturation with size) is weight-robust across the sweeps we ran. Finally, we exclude the ransom-payment game by design; CRADLE models the resilience of a restore-not-pay defender.

IX. CONCLUSION

We presented CRADLE, a transient stochastic-reward-net model of a crypto-ransomware double-extortion kill-chain

on a MEC micro-datacenter, defended by a footprint-aware endpoint monitor and an immutable-backup isolate-and-restore response. Unlike prior edge-under-attack models, CRADLE treats encryption as a recoverable service-denial state rather than an absorbing breach, orders exfiltration before encryption to capture double extortion, ties the detection hazard to the encryption and exfiltration footprint, and endows the response with explicit recovery-point and recovery-time economics. From one generator we obtained transient stage and service-denial probabilities, accumulated-reward losses in confidentiality, availability and data recency, first-passage detection and resolution distributions, and closed per-incident expectations via the fundamental matrix. We proved and demonstrated that coupling detection to attack progress makes the resolution law non-memoryless, a qualitative departure from uniform-recovery models. A co-design procedure jointly tuned detection sensitivity and backup cadence to minimize expected total cost, revealing a Pareto frontier and the substitutability of the two levers—so that sharper detection lets a defender back up less often—while showing that per-incident harm saturates with fleet size. An independent exact simulation validated every analytic result. Future work will lift the exponential assumption to semi-Markov sojourns for heavy-tailed encryption and restore times, embed an adaptive, state-triggered response policy, extend the model to federations of interacting micro-datacenters, and incorporate the strategic ransom-payment decision as a game against the modeled defender.

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